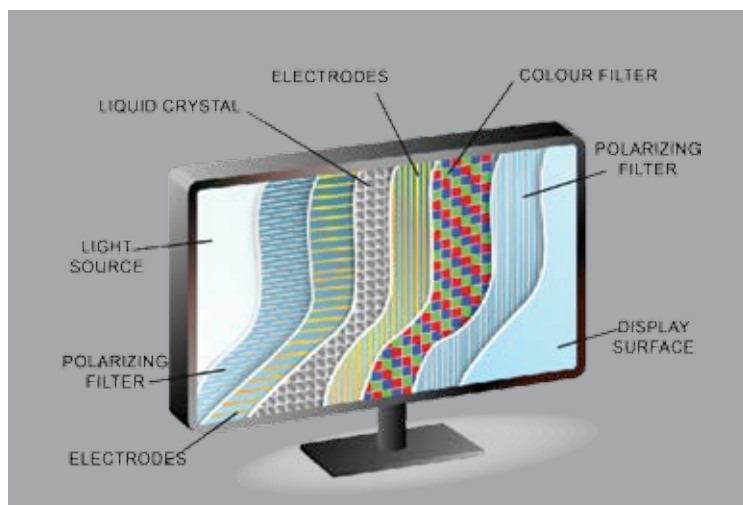


viously uncharged gas now consists of positively and negatively charged particles continuously bumping into each other and emitting ultraviolet light photons while they do so. These photons are then reabsorbed by the phosphorous in the cells. When an ultraviolet photon hits a phosphor atom in the cell, one of the phosphor's electrons jumps to a higher energy level and the atom heats up. When the electron falls back to its normal level, it releases energy in the form of a visible light photon.

The main advantage of plasma screens was the contrast ratio, considered one the best in the industry. Because plasma is able to provide deeper blacks and brighter whites, it has a high colour gamut. In addition, better viewing angles and lower backlighting problems due to the absence of a backlight made it a runaway hit. Alas, as enticing as this technology sounds, it has been discontinued by the biggies like Samsung and Panasonic. A big reason is burn-in due to static images, caused by the phosphors wearing out, further leading to a shorter life span. And since a plasma screen involves the use of a heavy electrode grid, it gets heavier for larger display sizes, with almost no corresponding increase in resolution.

Liquid Crystal Displays (LCDs)

The plasma display was followed up by the Liquid Crystal Display. It works with liquid crystals, which have intermediate properties between solid and



Cross section of a typical